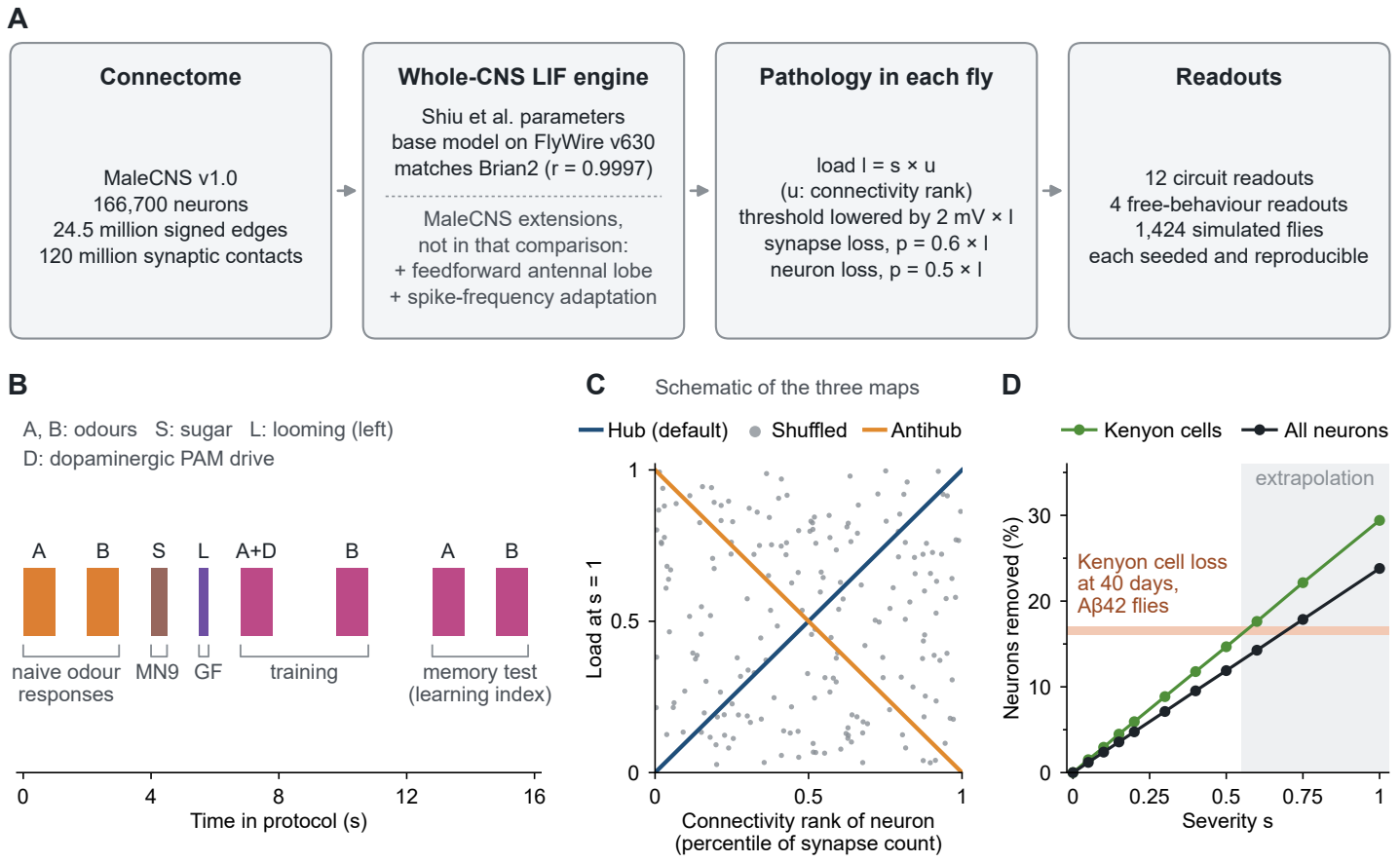

A
Connectome
MaleCNS v1.0
166,700 neurons
24.5 million signed edges
120 million synaptic contacts
Whole-CNS LIF engine
Shiu et al. parameters
base model on FlyWire v630
matches Brian2 (r = 0.9997)
MaleCNS extensions,
not in that comparison:
+ feedforward antennal lobe
+ spike-frequency adaptation
Pathology in each fly
load l = s × u
(u: connectivity rank)
threshold lowered by 2 mV × l
synapse loss, p = 0.6 × l
neuron loss, p = 0.5 × l
Readouts
12 circuit readouts
4 free-behaviour readouts
1,424 simulated flies
each seeded and reproducible
B
A, B: odours S: sugar L: looming (left)
D: dopaminergic PAM drive
A
B
S
L
A+D
B
A
B
naive odour responses
MN9
GF
training
memory test (learning index)
0
4
8
12
16
Time in protocol (s)
C
Schematic of the three maps
Hub (default)
Shuffled
Antihub
Load at s = 1
1
0.5
0
0
0.5
1
Connectivity rank of neuron (percentile of synapse count)
D
Kenyon cells
All neurons
extrapolation
Kenyon cell loss at 40 days, Aβ42 flies
Neurons removed (%)
30
20
10
0
0
0.25
0.5
0.75
1
Severity s